



Weather Data Analysis Using Data Distribution and Correlation Techniques



Exploratory Data Analysis • Statistical Distribution • Correlation-Based Insights

NAME: BHARATH M, REG NO: 192324302, COURSE : DSA 0405-FUNDAMENTAL OF DATA SCIENCE

Introduction

PROBLEM

How can weather variables be summarized and related?

APPROACH

Use distributions, summary statistics and correlation.

WHY?

Reveal patterns, variability and strong relationships.

Method Pipeline

Raw Weather Data

Data Cleaning

Distribution Analysis

Correlation Analysis

Insights

Assessments: mean, median, standard deviation, range, skewness, Pearson correlation (r), and visual pattern inspection.

Conclusion

1. Distribution analysis highlights the center and variability of weather measurements.
2. Correlation analysis identifies meaningful positive or negative relationships between variables.
3. Combining both techniques supports clearer exploratory weather interpretation.

Future Directions

- Add time-series forecasting and seasonal analysis.
- Compare multiple locations and weather stations.
- Apply anomaly detection and predictive models.

Dataset

Illustrative Weather Dataset

120 observations • Temperature • Humidity • Pressure • Wind Speed • Rainfall

120

Observations

5

Variables

°C

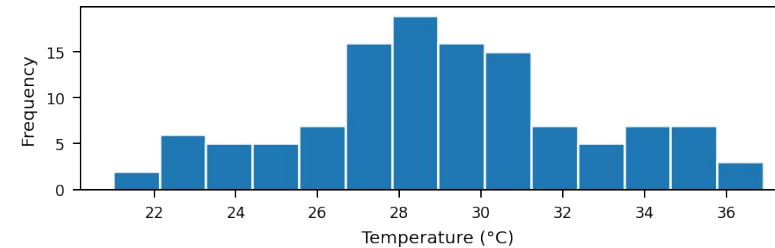
Temp. unit

mm

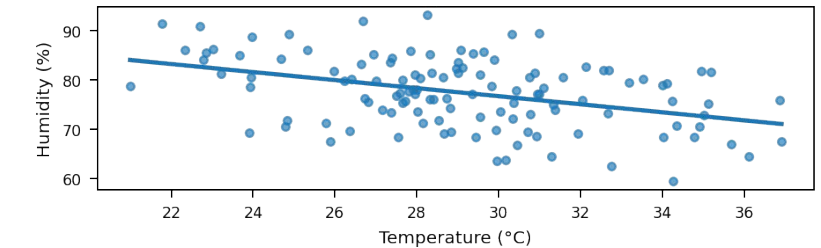
Rainfall

Results

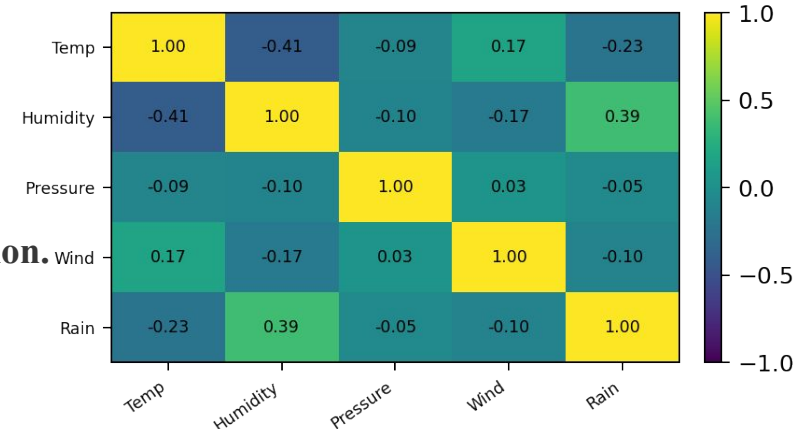
Temperature Distribution



Temperature vs Humidity (r = -0.41)



Correlation Matrix



Key Statistical Insights

- Temp–Humidity: $r = -0.41$
- Pressure–Wind: $r = 0.03$
- Humidity–Rainfall: $r = 0.39$
- Correlation indicates association, not causation.

Data preparation → Distribution → Correlation → Visualization → Interpretation